

Introduction to Modeling for Formal Verification

Today's goals

Module 1

Introduction to Model Checking

Module 2

Modeling

- Background
- Propositional Logic
Video • 13 min
- Predicate Logic
Video • 9 min
- Set Theory
Video • 11 min
- Foundations of Logic and Set Theory - Key Concepts and Applications
Reading • 1h
- Background Quiz
Graded Assignment • Grade: 100%

- Transition Systems
- Transition Systems: Definitions
Video • 13 min
- Deterministic vs Non-Deterministic Systems
Video • 8 min
- Executions
Video • 5 min
- Modeling Sequential Circuits
Video • 11 min
- Principles of Model Checking - Section 2.1, Page 19-29
Reading • 45 min
- Transition Systems Quiz
Graded Assignment • 20 min

- Program Graphs
- Program Graphs: Definitions
Video • 9 min
- Program Graphs: Examples
Video • 15 min
- From Program Graphs to Transition Systems
Video • 11 min
- Principles of Model Checking - Section 2.1, (Page 29-34)
Reading • 30 min
- Program Graphs Quiz
Graded Assignment • 15 min

Module 3

Modeling System Composition

- Modeling Concurrency Using Transition Systems
- Concurrency and Interleaving
Video • 10 min
- Concurrency and Interleaving: Examples
Video • 8 min
- Principles of Model Checking - Section 2.2.1, (Page 35-39)
Reading • 30 min
- Modeling Concurrency Using Transition Systems Quiz
Graded Assignment • 15 min

- Modeling Communications Via Shared Variables
- Interleaving Program Graphs
Video • 11 min
- Critical Actions and Atomicity
Video • 10 min
- Peterson's Mutual Exclusion Algorithm
Video • 10 min

Principles of Model Checking - Section 2.1, Page 19-29

0:13 / 0:37

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Textbook - Principles of Model Checking - Chapter 2.1 (page 19 - 29)

Definitions for transition systems, including predecessors, successors and examples, can be found on pages 19-23. The definition of deterministic transition systems is on page 24, and the discussions of nondeterminism can be found in the last paragraph on page 22. The concepts of executions and reachable states are explained in section 2.1.1. The first part of section 2.1.2, on pages 26-29, explains how sequential hardware circuits are modeled in transition systems.

Go to next item

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